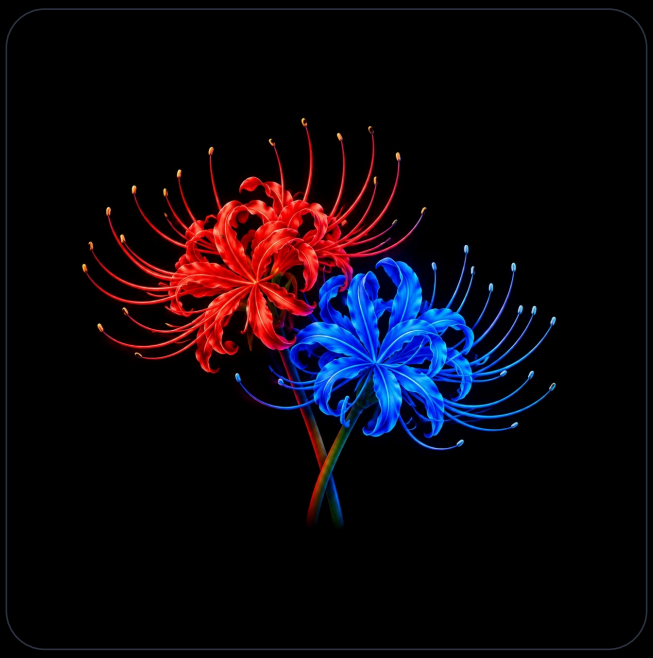


Spider Lily



Lotus Pond

SPIDER LILY & LOTUS POND

Gesture-Controlled Interactive 3D Web Apps

Bloom a glowing spider lily or a pond of twelve lotuses with nothing but your hands - tracked live through your webcam, running entirely in the browser.

Computer Vision - AI / ML - 3D Modeling - Web - Interaction Logic

bhonepyaemin.com/spiderlily

bhonepyaemin.com/lotus

Source: github.com/BhonePyaeMin/Spider-Lily - github.com/BhonePyaeMin/Lotus

About the project

Spider Lily and Lotus Pond are two interactive pieces of generative 3D art that share one idea: your hand is the controller. Hold it up to the webcam and the flowers respond in real time - spread your thumb and index finger and they bloom, bring them together and they close back into buds.

Both run as a single self-contained web page with no build step and no server-side code. Hand tracking runs on your own device, so the camera video is never uploaded anywhere. Spider Lily began as a TouchDesigner piece and was rebuilt for the browser; Lotus Pond extends the same engine to a whole pond.

Two scenes

Spider Lily

A procedural spider lily (Lycoris) with long, sweeping stamens on a black stage with a soft bloom glow. Each flower head is an umbel of florets, and the bloom plays as a staged choreography: the bud loosens, petals lift and separate, arch outward and curl back, then the stamens extend.

Lotus Pond

Twelve lotus flowers, each on its own stem with its own colour variety. They open in a staggered order through three growth stages - Bud Formation, Vegetative and Anthesis - so the pond always shows buds and full blooms at the same time.

How you control them

Spread thumb + index

Blooms the flowers. The gap between your fingertips is measured relative to your hand size, so it works at any distance from the camera. Either hand works - the widest spread wins.

Pinch fingers together

Closes the flowers back into buds. Hold any gap to freeze the bloom at that stage.

Move an open hand

The flowers follow your hand across the screen while you keep blooming them. One or two hands both work.

Make a fist

Freezes the bloom (a fist would otherwise read as a pinch) while you keep moving the flowers.

Keyboard and mouse

D toggles a demo bloom cycle with no camera needed; 1 / 2 / 3 hold the bloom at bud, midway or full; 0 resumes the cycle; H hides the status overlay. Drag or scroll to orbit and zoom the camera.

Inside each scene

Spider Lily - the bloom choreography

The bloom is an anime.js timeline of key poses - tight bud, lift, arch, full bloom, settle. Your fingers scrub that timeline back and forth, so the stages always play in order, forwards or in reverse, in step with your hand. Stems are built recursively, and every stamen is its own curved filament.

Lotus Pond - procedural petals

Thick, cupped, boat-shaped petals in four layers around a domed seed pod dotted with seed bumps and ringed with stamens. Closed flowers form pointed teardrop buds, and each flower opens on its own staggered schedule.

Lotus Pond - natural rendering

Soft shadows between overlapping petals, baked ambient occlusion and fine lengthwise veins in the petal colours, a velvety sheen on the petal surface and a gentle glow on the edges.

Technology stack

Computer Vision

Real-time webcam capture (getUserMedia) with hand detection on every video frame and a live landmark overlay drawn on the camera feed.

AI / Machine Learning

MediaPipe HandLandmarker - a pretrained neural network that returns 21 landmarks per hand for up to two hands, running on-device via WebAssembly with GPU acceleration and a CPU fallback.

Web

JavaScript (ES modules), HTML5 and CSS3 in a single index.html per scene. Three.js 0.166 renders with WebGL, with EffectComposer and UnrealBloomPass for the glow and OrbitControls for the camera.

Animation

anime.js 3.2 timelines for the staged bloom, scrubbed directly by hand input with light time-based smoothing to absorb tracking jitter.

Interaction Logic

Hand-size-normalised thumb-to-index distance, palm tracking mapped to world space along the camera axes, fist detection, handedness mapping and jitter dead-zones.

3D Modeling

All geometry is generated in code: recursive stems, petal blades rebuilt every frame with recurve and ripple, cupped multi-layer lotus petals, seed pods, stamens and layered vertex colouring.

How it works

1 **Webcam**
getUserMedia streams the camera into a video element, shown mirrored like a selfie view.

2 **Hand tracking**
MediaPipe HandLandmarker finds up to two hands and returns 21 landmarks each.

3 **Gesture logic**
Fingertip gap, palm position and fist state become bloom and movement values.

4 **Smoothing**
A time-based lag filter absorbs tracking jitter without feeling laggy.

5 **3D render**
Three.js poses the petals and stamens, moves the flowers and draws them with a glow pass.

Try them

Spider Lily

Live demo: bhonepyaemin.com/spiderlily
Source: github.com/BhonePyaeMin/Spider-Lily

Lotus Pond

Live demo: bhonepyaemin.com/lotus
Source: github.com/BhonePyaeMin/Lotus

Open either demo in Chrome or Safari and allow the camera - browsers only allow camera access over HTTPS or localhost. To run one locally, serve its folder (for example python -m http.server) and open it on localhost; there is nothing to install or build.

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